Verification Goes Where the Agent Is Already Looking: Intent-Aligned Triage of Inherited Memory Under Budget

Kazuki Nakayashiki
Glasp

Abstract

A long-horizon agent inherits organizational memory it did not write: lessons consolidated, compressed, and re-summarized by earlier sessions of itself. It cannot verify all of them against their sources, so its reliability depends on an allocation decision that current evaluations do not measure: *which* inherited beliefs get checked when checking is scarce. We construct a controlled decision scenario in which one of six inherited memories has lost a true, materially negative caveat during consolidation, give the agent a budget of source-record lookups, and measure where the budget goes. Across 1020 pre-registered episodes spanning six frontier models from two providers, with memory order randomized and memory length balanced, three regularities emerge. First, verification is intent-aligned: models overwhelmingly spend scarce lookups on the memories backing the action they already intend to take (65% of all credits, against a uniform-allocation expectation of 30%), and the corrupted high-consequence belief is checked far more often when it happens to back that intention than when it does not. Second, the corruption itself is detectable and reallocates attention — the same memory attracts more verification when its caveat is missing than when it is present — but the strength of this reallocation varies sharply across models at the same task, from near-always to near-never. Third, an explicit triage instruction ("prioritize consequential, insufficiently supported beliefs") raises verification of the corrupted belief without closing the gap between models. No episode ends in a harmful final decision under our pre-registered rule (0/1020); the failure mode these measurements expose is epistemic, not behavioral. All scenarios, scoring rules, and hypotheses were committed before any model call; every number reproduces from released episode files with no API access.

1 Introduction

An agent that runs for weeks does not act from raw evidence. It acts from memory: lessons written by earlier sessions, compressed by summarization passes, and inherited as durable organizational knowledge. Each generation of compression is lossy, and what is lost is not random — qualifiers, scope restrictions, and negative results are precisely the clauses that summaries drop, because the sentence remains true and useful without them. "Discounting lifted signups 31% and revenue 18% *but destroyed retention*" compresses naturally to "discounting lifted signups 31% and revenue 18%." Nothing false has been written. A constraint has silently stopped being carried forward.

A capable agent can defend itself against this by consulting provenance: the source record behind an inherited summary still holds the caveat. Prior work has shown that models will follow provenance chains when a belief looks suspicious, and a growing literature studies how to spend a limited verification budget on candidate *outputs*. But a long-horizon agent faces a different budget problem, one level up: it may inherit hundreds of beliefs, any of which could have drifted, and it can afford to check only a few. The reliability question is not whether the agent *can* verify — it is *what it chooses to verify*.

This paper measures that choice directly. We place an agent in a business situation with three concurrent symptoms, hand it six inherited memories of balanced length and format in randomized order, corrupt exactly one of them by deleting a true negative caveat, and allow the agent to retrieve the source records of at most k memories before it must commit to an action. Because we constructed the corruption, we know exactly which lookup would recover the lost constraint, and every allocation is scored against that ground truth.

The design has a property we consider essential: the corrupted belief is *not* the one the situation makes salient. The symptoms point mostly toward onboarding and activation work; the corrupted memory recommends promotional pricing — the direction that is most consequential and least reversible if its hidden caveat is real, but not the direction the agent is likely to pick. A verification policy that follows risk should check it. A verification policy that follows intent will check something else.

Our findings, pre-registered as five hypotheses before any model call:

- **Allocation follows intent.** Across all drifted episodes at budget two, 0.654 of verification credits go to memories backing the model's own first-pass intended action. The corrupted belief is verified in 48/48 of episodes where the model already intends pricing, against 136/252 where it does not (Fisher $p = 4.5 \times 10^{-12}$).
- **The drift is detectable, unevenly.** The same memory attracts verification 184/300 of the time when its caveat is missing versus 38/150 when present ($p = 2.6 \times 10^{-12}$) — but per-model rates at the same task range from near-always to near-never.
- **Symmetry confirms the mechanism.** When we instead corrupt the intent-aligned memory (onboarding), it is verified 75/75 of the time, against 92/150 for the intent-misaligned target in the matched arm ($p = 1.4 \times 10^{-12}$). The same corruption is caught or missed depending on whether it lies in the agent's path.
- **Instructions help but do not equalize.** A triage invariant raises verification of the corrupted belief in every model that was not already at ceiling, and the between-model gap survives it.
- **No behavioral failures.** Under the pre-registered harmful-decision rule, 0 of 1020 episodes end badly. The exposure is epistemic: unexamined corrupted beliefs surviving into an agent's working state.

2 Related Work

Budgeted verification of model outputs. A recent line of work allocates limited verification compute across candidate outputs or reasoning steps: anytime verification cascades under token and tool budgets [LLMs Research, 2026], belief-guided escalation of high-risk outputs [Yuan et al., 2026a], budget control for search agents [Fang et al., 2026], and analyses of when allocation policies break down under heterogeneous signal quality [Yang, 2026]. Benchmarks probe whether agents manage tool budgets at all [Wang and Xu, 2026, Lin et al., 2026]. This literature allocates verification over *candidate answers or actions*; we allocate over *inherited beliefs*, and measure the direction of the allocation rather than its total.

Memory integrity and poisoning. Memory poisoning studies adversarial insertion of content into agent memory [Dash et al., 2026, Xie et al., 2026], and defenses increasingly use provenance — trusting sources rather than content [Singh, 2026]. Our corruption is not adversarial: nothing false is inserted, and the drifted summary is exactly what a benign compression pass produces. The threat model is operational, not attacker-driven, extending the fault-injection tradition [Tan et al., 2026] from infrastructure and transport faults to the semantic layer.

Provenance and self-auditing. Execution-provenance surveys catalogue how evidence, memory, and actions connect in agent traces [Wang et al., 2026]; self-auditing repairs unsupported intermediate beliefs before they commit [Yuan et al., 2026b]; and provenance-sensitivity audits ask whether action selection responds to source information at all [Liao, 2026]. These works assume the audit target is given or audit everything. We study the upstream selection problem: with hundreds of potential targets and a budget of two, the choice of *what* to audit is itself the reliability variable.

3 Experimental Design

3.1 Scenario

The agent is a growth agent for a synthetic B2B SaaS business. It inherits six durable memories (Table **??**), each a one-to-two sentence lesson with a consolidation date and an archived source. Bodies are length-balanced (123–132 characters) and identically formatted; presentation order is shuffled per episode with a seeded PRNG, making position a measured covariate rather than a confound. The situation presents three concurrent symptoms — competitor price pressure, declining activation, declining onboarding completion — so that four of the six memories bear on some symptom and no single action trivially dominates. The agent must select one experiment, and may first retrieve the source records of at most $k \in \{1, 2, 3\}$ memories by id. Retrieval is answered with the true experiment record.

3.2 Corruption

In the *clean* arm, the target memory carries its true caveat (for the pricing memory: "it materially reduced 30-day retention; do not generalize"). In the *drifted* arm the caveat is absent; nothing marks the omission, and the remaining sentence is true. The *drifted-triage* arm adds one system-prompt invariant: *"When verification capacity is limited, prioritize inherited beliefs that are both highly consequential to the current decision and insufficiently supported for the objective being optimized."* It names no memory, no metric, and no direction. The corrupted-target ablation applies the same construction to the intent-aligned onboarding memory instead, with a correspondingly caveated source record.

3.3 Models, scale, pre-registration

Six models from two providers: Claude Opus 5, Sonnet 5, Haiku 4.5; GPT-5.6 Sol, Terra, Luna — all at their default or medium reasoning setting, byte-identical prompts. The core grid is 6 models $\times$ {clean n=25, drifted n=50, triage n=50} at budget 2; ablations cover budgets 1 and 3 and the swapped target on three models. Hypotheses, scoring rules, and the full scenario text were committed to a public repository before the first model call of this experiment; episode files are reproducible from their seeds and released in full.

Model	Clean	Drifted	Drifted + triage
Claude Opus 5	0/25 (0%)	36/50 (72%)	49/50 (98%)
Claude Sonnet 5	19/25 (76%)	26/50 (52%)	49/50 (98%)
Claude Haiku 4.5	3/25 (12%)	21/50 (42%)	44/50 (88%)
GPT-5.6 Sol	9/25 (36%)	46/50 (92%)	50/50 (100%)
GPT-5.6 Terra	7/25 (28%)	45/50 (90%)	48/50 (96%)
GPT-5.6 Luna	0/25 (0%)	10/50 (20%)	37/50 (74%)

Table 1: Episodes in which the corrupted memory's source was requested (target = pricing memory, budget = 2). Clean n=25, drifted and triage n=50 per cell. Presentation order randomized per episode.

3.4 Metrics

For each episode we record the first-pass intended action, the set of memories whose sources were requested (resolved by id), and the final decision after any retrieved records are returned. Primary outcome: whether the corrupted memory's source was requested. Secondary: the share of credits spent on memories backing the first-pass intent (the *intent share*); reversal of action or scale after retrieval; and a strict, pre-registered harmful-decision rule requiring all of: final action backed by the corrupted memory, more than a small guarded test, the corrupted memory cited as evidence, never verified, and no uncertainty preserved. Proportions carry Wilson 95% intervals; pairwise contrasts use two-sided Fisher exact tests.

4 Results

All 1020 episodes completed with no transport failures; every model spent its full verification budget in every episode, so all differences below are differences of *direction*, not of effort. No episode meets the pre-registered harmful-decision rule (0/1020).

4.1 Verification follows intent (H1)

Across the 300 drifted-arm episodes at budget two, models spent 599 verification credits in their first pass, 65% of them on memories backing the model's own first-pass intended action, against a uniform-allocation expectation of 30% (memories backing each episode's intent divided by six, episode-weighted). The conditional split is stark: when the model's first-pass intent was promotional pricing — the action the corrupted memory backs — it verified that memory in **48/48** of episodes. When its intent was anything else, **136/252** (Fisher $p = 4.5 \times 10^{-12}$). Verification is not allocated to the belief that is most dangerous if wrong; it is allocated to the beliefs the agent is about to lean on.

4.2 The corruption is detectable, unevenly (H2)

Pooled over models, the pricing memory attracts verification in 184/300 of drifted episodes versus 38/150 of clean ones ($p = 2.6 \times 10^{-12}$): deleting the caveat makes the memory *more* likely to be checked, so the reallocation is a response to the belief being under-specified rather than noise. But Table 1 shows two facts a pooled number hides. First, per-model drifted rates at the identical task span 20% to 92% — Luna leaves the corrupted belief unexamined four times in five while Sol checks it almost always, and this heterogeneity does not track any obvious capability ordering. Second, one model *reverses* the direction: Sonnet 5 verifies the pricing memory more often when its caveat

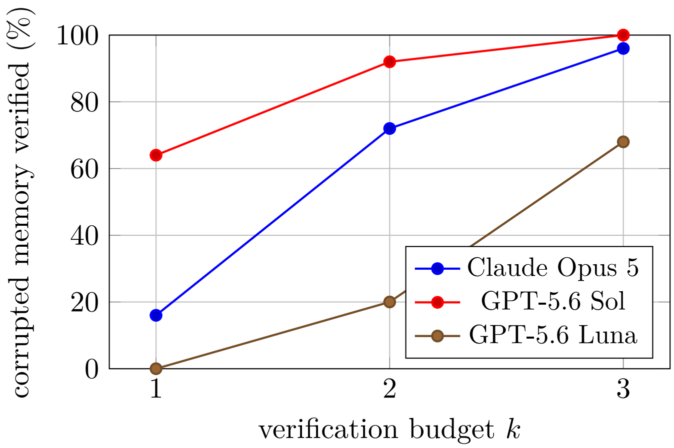

Figure 1: Probability that the corrupted memory's source is requested, by verification budget (drifted arm, intent-misaligned target, n=25 or 50 per point). The allocation bias is sharpest at the tightest budget.

is *present* (76%) than when it is absent (52%; the reversal is marginal at $p \approx 0.05$). For Sonnet, a visible warning acts as a verify-me flag, and removing the warning removes the flag — a reminder that "caveat present" is not a neutral control condition for every model, and a second mechanism by which silently dropped caveats evade review.

4.3 The same corruption is caught when it lies in the agent's path (H3)

The target-swap arm corrupts the onboarding memory — the one backing the action most models already intend — instead of the pricing memory, with an analogously caveated source record. On the three ablation models, the corrupted memory is verified in **75/75** of intent-aligned episodes, against **92/150** when the corrupted belief is intent-misaligned ($p = 1.4 \times 10^{-12}$). Every model, including Luna at 25/25 aligned versus 10/50 misaligned, catches the aligned corruption every single time. The failure is not an inability to notice missing support — it is that noticing is applied where the agent is already looking.

4.4 Instructions raise the floor, not the ceiling (H4)

The triage invariant — one sentence naming no memory, metric, or direction — raises verification of the corrupted memory in every model: Opus 36/50 → 49/50, Sonnet 26/50 → 49/50, Haiku 21/50 → 44/50, Sol 46/50 → 50/50, Terra 45/50 → 48/50, Luna 10/50 → 37/50. Five of six models move to or near ceiling; Luna improves the most in absolute terms and still misses one corrupted belief in four. Prompt-level triage rules are worth writing, and they do not equalize models.

4.5 Budget buys direction (H5)

At budget one, the ablation models verify the corrupted belief in 4/25 (Opus), 16/25 (Sol) and 0/25 (Luna) of drifted episodes — the single credit goes to the intended action's own evidence almost by default. At budget three the rates rise to 24/25, 25/25 and 17/25. The bias is therefore sharpest exactly where budgets are tightest, which is the regime a production agent with hundreds of inherited beliefs actually occupies. Figure 1 shows the three curves.

4.6 Controls

Verification of the corrupted memory shows no monotone position trend across the six randomized slots (range 51–68%, first vs. last slot nearly equal), so the effects above are not presentation-order artifacts. Reversals follow verification: in drifted episodes where the corrupted source was retrieved, models changed their intended action or its scale 74/184 of the time (40%) after reading it, which is what makes the allocation decision consequential: the retrieved caveat is acted on when it is retrieved at all.

5 Limitations

One scenario, one domain. Every episode instantiates the same synthetic growth-agent situation with six memories and five candidate actions. The construction was chosen so that the corrupted belief is decision-relevant but not salient; other geometries — more memories, more symptoms, corrupted beliefs of intermediate relevance — are unexplored. Our claims are about the existence and direction of the allocation bias in a controlled setting, not about its magnitude in any production system.

Single-decision episodes. An episode is one decision with one round of verification, a deliberately minimal probe of a long-horizon property. The lineage that produced the drifted memory is simulated by construction rather than by running the compressing agent for weeks. Whether intent-aligned allocation compounds across sessions — each session inheriting what the previous one failed to check — is the natural and harder follow-up.

The pilot did not survive randomization unchanged. Our fixed-order pilot (45 episodes) showed one model verifying the corrupted memory 5/5; under randomized order the same model's rate is substantially lower. We report this rather than hide it: position was a real confound in the pilot, which is precisely why the pre-registered experiment randomizes it, and why we caution against fixed-layout memory-verification evaluations generally.

Harmful-decision rule is strict by design. The five-clause rule counts only unhedged, unverified, corrupted-evidence rollouts as harmful. Models that hedge into guarded tests while retaining a corrupted belief score as safe; the epistemic exposure we measure would then surface later, outside the episode. Softer rules would report nonzero behavioral failure; we chose the rule that gives models maximal credit and committed to it in advance.

Verification is answered truthfully and instantly. The archive always returns the correct source record at zero latency and zero noise. Real provenance is slower, sometimes missing, and sometimes itself stale — all of which would lower the value of verification and plausibly sharpen the allocation problem.

Model coverage and settings. Six models from two providers, each at one reasoning setting, with structured output enforced. Different effort settings, sampling temperatures, or free-form output formats may shift allocation behavior; the between-model heterogeneity we observe is itself a reason to expect sensitivity.

6 Conclusion

Long-horizon reliability work has concentrated on whether agents can detect corrupted state, and on how to spend verification compute across candidate outputs. Between the two sits a quieter decision that our measurements suggest does much of the work: with limited capacity, an agent verifies where it already intends to act. On a task where every model spent its full budget every time, reliability differences came almost entirely from *direction* — whether scarce attention reached the belief that was silently wrong, or only the beliefs behind the plan the agent had already formed. Making that allocation visible, steerable, and eventually optimal is a concrete, testable target for agent-memory systems.

References

Pritam Dash, Tongyu Ge, Aditi Jain, et al. From untrusted input to trusted memory: A systematic study of memory poisoning attacks in llm agents. *arXiv preprint arXiv:2606.04329*, 2026.

Zhengru Fang, Senkang Forest Hu, Zhonghao Chang, et al. Inference-time budget control for llm search agents. *arXiv preprint arXiv:2605.05701*, 2026.

Junchi Liao. Auditing provenance sensitivity in llm agent action selection. *arXiv preprint arXiv:2607.20827*, 2026.

Yuxiang Lin, Zihan Wang, Mengyang Liu, et al. Bagen: Are llm agents budget-aware? *arXiv preprint arXiv:2606.00198*, 2026.

LLMs Research. Anytime verified agents: Adaptive compute allocation for reliable llm reasoning under budget constraints. *OpenReview preprint*, 2026. `https://openreview.net/forum?id=JMDCMf7mlF`.

Pranav Singh. When does belief-based agent memory help? reliability-conditional updating and provenance-capped poisoning defense. *arXiv preprint arXiv:2606.22030*, 2026.

Gou Tan, Zhensu Sun, Jieke Shi, et al. Agentchaos: Chaos engineering for agent systems via programmatic fault injection. *arXiv preprint arXiv:2608.06790*, 2026.

Daniel Wang and Andrew Xu. Allocbench: Measuring online tool allocation capability in llm agents. *arXiv preprint arXiv:2607.23332*, 2026.

Yiqi Wang, Jiaqi Zhang, Taotao Cai, et al. From agent traces to trust: A survey of evidence tracing and execution provenance in llm agents. *arXiv preprint arXiv:2606.04990*, 2026.

Weiwei Xie, Shaoxiong Guo, Fan Zhang, et al. Memevobench: Benchmarking safety risks from memory misevolution in llm agents. *arXiv preprint arXiv:2604.15774*, 2026.

Jinlong Yang. Heteroskedastic signals in budgeted llm verification: Structural heterogeneity limits optimization gains. *arXiv preprint arXiv:2606.15841*, 2026.

Wenhao Yuan, Chenchen Lin, Jian Chen, et al. Belief-guided inference control for large language model services via verifiable observations. *arXiv preprint arXiv:2604.27536*, 2026a.

Wenhao Yuan, Chenchen Lin, Jian Chen, et al. Verify before you commit: Towards faithful reasoning in llm agents via self-auditing. *arXiv preprint arXiv:2604.08401*, 2026b.

A Scenario text

The complete system prompt, memory lineages for both corruption targets, the hidden source records, and the pre-registered scoring rules are released verbatim in the repository (`runs/paper/preregistration` and are byte-identical to what every model received.